

Shrodinger: A Techno-Cognitive Architecture for Active Learning

Quantum Cats

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Abstract

This document outlines the architecture of *Shrodinger*, a dynamic reading infrastructure designed to counteract **Eduslop**. By integrating **LangChain** orchestration, vectorized knowledge tracing, and high-fidelity auditory feedback, Shrodinger transforms source documents into interactive interfaces that promote **critical thinking** through productive friction.

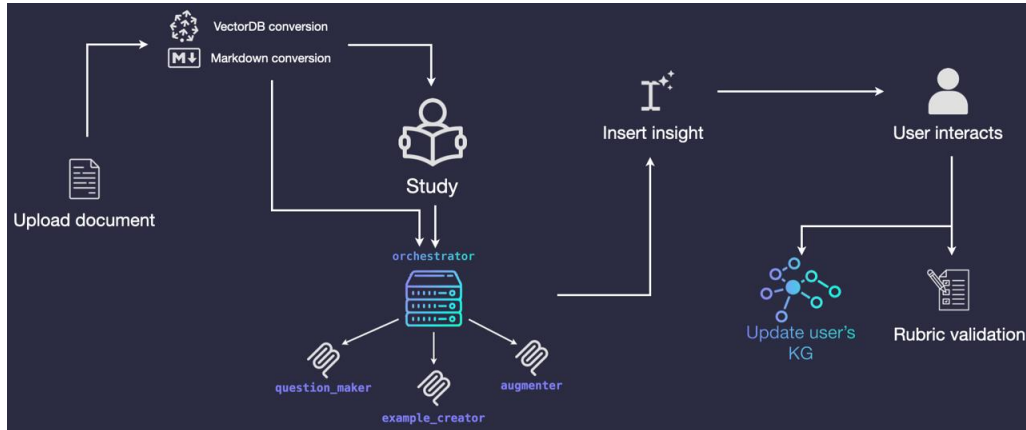


Figure 1: High-Level Workflow: From Ingestion to Dynamic Study Loop.

1 Vision: Overcoming Eduslop via Productive Friction

The current AI-EdTech landscape suffers from a systemic pathology: confusing algorithmic output with the human learning process. When an AI provides pre-packaged answers, it generates **Eduslop**, favoring **Cognitive Offloading**. Our approach is based on the principle of **Productive Friction**: learning occurs only when the learner overcomes a desirable difficulty. The Shrodinger architecture implements **Scaffolding**, building a dynamic infrastructure that supports meaning construction before dissolving as competence is acquired.

2 Source-Driven Paradigm and Dynamic Injection

Unlike *Chat-driven* systems, Shrodinger adopts a **Source-driven** interface. The original document is the un-editable interface; the AI executes its mutations within the text. Technically, we utilize **React** and **Tiptap** (**ProseMirror**) for rendering. Tiptap transforms the source into a manipulatable Abstract Syntax Tree (AST). Through custom *Node Views*, the system injects interactive components directly into the textual coordinates.

3 Dual Vector Knowledge Tracing

To ensure real-time response, Shrodinger implements a **Dual Vector Space** architecture managed by an asynchronous **FastAPI** server. Chunks are processed via embedding models to constitute the **Source Vector DB**, while a **User Vector DB** is initialized to map the student's knowledge. FastAPI executes a **cosine similarity** calculation to determine the mathematical delta between the text's requirements and the user's state.

4 Multi-Agent Swarm and LangChain Orchestration

To manage the complexity of concurrent agent interactions, we utilized **LangChain** as the primary orchestration framework. The backend invokes a framework of specialized agents (**The Swarm**) responsible for generating questions, examples, and curiosities. To prevent hallucinations and ensure UI stability, we strictly enforce **Structured Outputs** using **Pydantic** schemas. This ensures that every piece of friction—from cloze tests to analogies—is delivered as a perfectly typed JSON object, ready for frontend injection.

5 Rubric-Based Evaluation and Outcomes

We reject chatbot discursive validation in favor of the *LLM-as-a-Judge* pattern. Student responses are decomposed and evaluated according to a formal **Rubric**. The AI acts as a binary judge, verifying the presence of key concepts and technical terms. This Rubric outcome mathematically updates the User Vector DB, closing the **Golden Loop** of adaptive learning without relying on unquantifiable narrative feedback.

6 Auditory Scaffolding: ElevenLabs Integration

To reduce visual cognitive load, Shrodinger integrates an auditory layer via the **ElevenLabs API**. This component acts as a "Voice Companion" providing narrative support and metacognitive feedback through high-fidelity synthetic voices. The integration uses **FastAPI** to stream text-to-speech data directly to the **React** frontend, ensuring near-zero latency.